

now identified as an automation server. Jenkins supports Gradle based Android projects.

Stable release	2.121.3
Licence	MIT
Website	https://jenkins.io
Written in	Java
On-premise or hosted	The open source version is available in different flavours for different operating systems to install on premise. CloudBees provides the enterprise version of Jenkins.
Main features	Easy installation. Role based access. Distributed architecture – Master agent. More than 400 plugins are available to integrate with different open source and commercial tools. Flexible UI. Supports DevOps practices such as continuous integration and continuous delivery.

Download Jenkins for Windows from <https://jenkins.io/download> and install it. Go to the Jenkins dashboard at <http://localhost:80808> and click on *Manage Jenkins*.

- Go to *Manage Plugins* and then to *Available Tab* and install the Gradle plugin.
- Verify the Gradle plugin installation in the *Installed* tab.
- Go to *Manage Jenkins > Configure System > Global properties > Add Environment Variable ANDROID_HOME* and provide the SDK path in the value.
- Go to *Manage Jenkins > Configure System > Global Tool Configuration > Configure Gradle installations*.
- Create the Freestyle Project from the Jenkins dashboard.
- Go to the *Source Code Management* section and configure the code repository details.
- Go to the *Build* section.
- Click on the *Add build* step and select *Invoke Gradle Script*.
- Click on *Save* and then on *Build now* to create an APK file for Android Project.

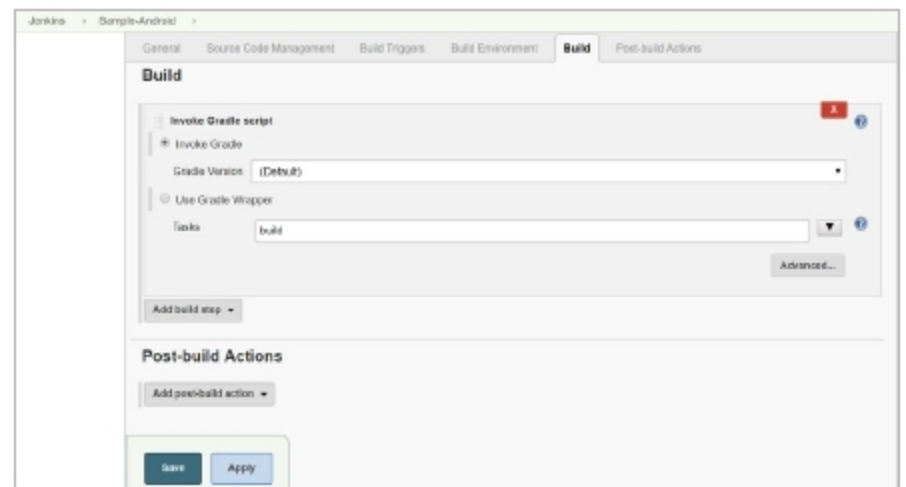


Figure 12: Jenkins: Invoke Gradle script

Continuous code inspection

Across organisations, different practices are followed for maintaining code quality and standards. The main reason to ensure a good quality of code is to make it maintainable and easy to understand. Manual verification of code is very difficult and may not give fruitful results. Different tools are used for code analysis for Android projects in Windows.

Android Lint

Android Studio comes with Android Lint. It verifies the quality of code and gives a list of errors and warnings.

- Create a Freestyle Project from the Jenkins Dashboard.
- Go to the *Source Code Management* section and configure code repository details.
- Go to the *Build* section.
- Click on the *Add build* step and select *Invoke Gradle Script*.
- Input *Lint* task.
- Click on *Save* and then on *Build now*. **END** 

References

- [1] Jenkins Essentials - Second Edition: Setting the stage for a DevOps culture <https://www.amazon.com/gp/product/B073PGCKHY>

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